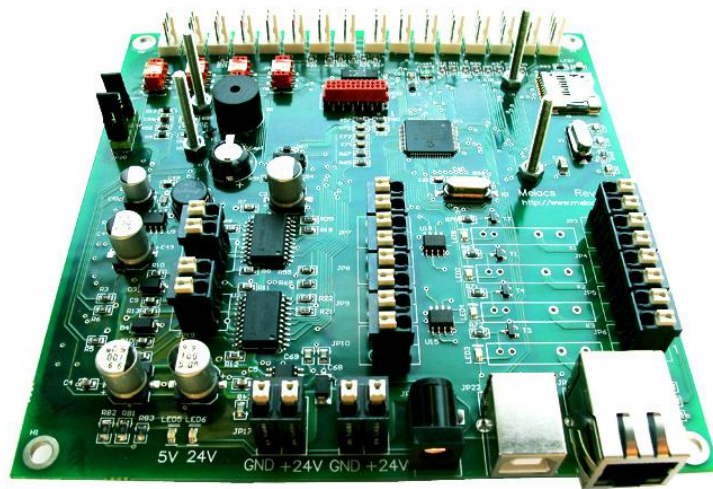


MELACS

Micro Energy Logger and Control System

Application Manual

General Edition — Control Platform for Thermal, Electrical and Energy-Storage Plants



Melacs control board — general-purpose hardware platform

Document status: general / application-independent edition
Revision: 1.0 — 2026

Notice

This manual describes the Melacs control platform as a general-purpose control system for energy plants. It replaces application-specific example content with a generic description of the hardware, inputs and outputs, communication interfaces, operating modes and status reporting that are common to all Melacs installations, independent of which plant equipment is connected.

The Melacs platform has been in continuous development since 2002 and is used across a range of thermal and electrical energy applications, including solar thermal collection, heat pumps, thermal energy storage, biofuel and biomass systems, power electronics and grid interfaces, and Power-to-X / e-fuel installations.

Information in this document is subject to change without notice as the platform and its firmware evolve. Application-specific behaviour (exact sensor assignments, setpoints, control sequences and status text) is defined separately for each installation and is documented in the corresponding application supplement.

All internally embedded platform software and control-board firmware, together with this manual, are proprietary. Application-specific configuration and control logic developed for a particular installation may be subject to separate rights held by the party responsible for that application.

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1. About This Manual

This is the general edition of the Melacs Application Manual. Earlier manuals were written around one specific installation (for example, a solar-tracking heating system) and mixed platform-level facts with application-specific detail. This edition separates the two: it documents what is true of every Melacs installation — the control board, its inputs and outputs, its communication interfaces, its operating modes and its status/alarm reporting — and leaves the plant-specific detail (which sensor is connected where, which setpoints apply, which sequence the valves follow) to a short application supplement written for each project.

Use this manual to understand how the Melacs platform is built and how to commission, operate and troubleshoot it at the platform level. Use the application supplement for the specific control strategy, sensor list and status-code meanings of an individual plant.

2. System Overview

Melacs ("micro energy logger and control system") is a compact embedded control platform for monitoring and controlling energy-conversion and energy-storage equipment. The same board and firmware base is used across heat, cold and electrical-energy applications; what changes between installations is the application program that reads sensors, evaluates conditions and drives outputs for the specific plant.

A Melacs control board typically performs three functions in an installation:

- Acquisition — reading analog sensors (temperatures, irradiance, flow, pressure, level) and digital status signals (inverter status, limit switches, external alarms).
- Control and logic — evaluating the acquired data against an application-specific algorithm and driving actuators (pumps, valves, dampers, contactors, position actuators) and setpoints to connected equipment (heat pumps, boilers, inverters, electrolyzers).
- Logging and communication — recording process data to local storage and making it available locally (display, USB) and over a network (Ethernet web server, RS-485/Modbus RTU, and where fitted, cellular/GSM).

Because acquisition, control and communication are handled by the same platform, Melacs can be configured as the primary controller of a small plant, or as a data-logging and interface node reporting into a larger site or fleet-level energy-management system.

3. Control System Architecture

The diagram below shows the general structure common to Melacs installations: sensor inputs on the left, actuator and setpoint outputs on the right, communication and logging interfaces at the bottom, and — purely as illustration — examples of the kind of plant equipment a Melacs board is typically connected to across different applications. Any single installation uses the subset that applies to its own plant.

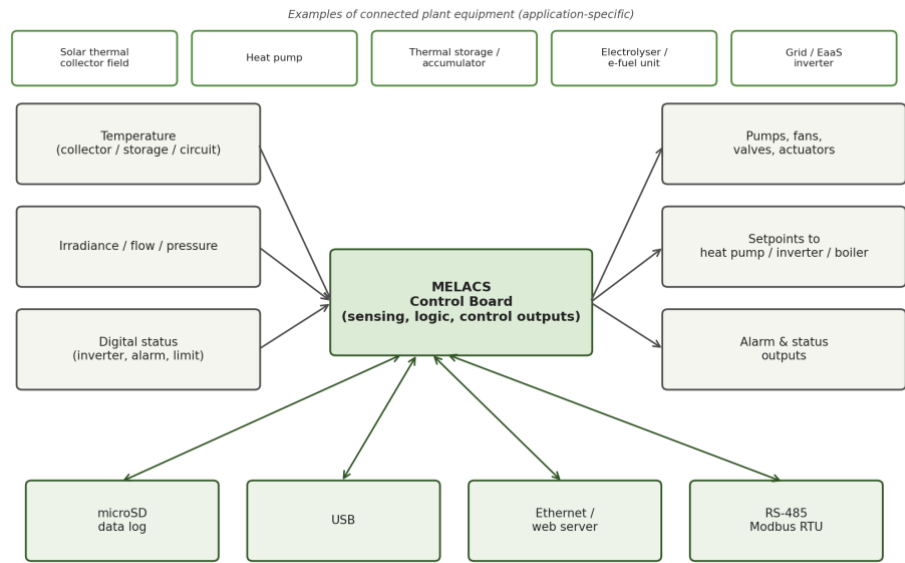


Figure 1 — General Melacs control system architecture (application equipment shown is illustrative)

3.1 Control board

The control board carries the processing core, analog and digital I/O circuitry, real-time clock, local display interface, microSD card socket, USB device port and Ethernet interface. An extension board can be added on the same communication bus to increase the number of available I/O channels for larger installations, and additional nodes (for example a grid inverter, a motor/actuator controller or a dedicated sensor board) can be attached over RS-485 for distributed installations.

3.2 Application software

The behaviour of a given installation is defined by an application program that runs on the control platform and by a configuration held on the plant's memory card. This separation is what allows the same board and base firmware to serve a solar-thermal system, a heat-pump installation, a thermal-storage/accumulator system, or a Power-to-X (electrolysis / e-fuel) unit: only the application layer changes.

4. Inputs and Outputs

Every Melacs control board exposes the same physical channel groups: analog inputs (AN-0 ... AN-7), digital inputs/outputs (IO-0 ... IO-7), and power/actuator channels (JP3 ... JP19) for driving pumps, valves and other loads directly or through the extension board. The tables below describe the generic function of each channel group. For a specific installation, the application supplement lists exactly which sensor or actuator is wired to which channel.

4.1 Analog inputs

Analog sensors are connected to the board's A/D channels. Channel numbers are marked on the board silkscreen.

Signal	Channel	Generic function	Type	Typical use / notes
1	AN-7	Irradiance / general analog input	Input	Solar irradiance sensor, or repurposed for another analog measurement
2	AN-6	Position / orientation input	Input	Inclinometer or position feedback, where an actuated component is used
3	AN-5	Free analog input	Input / output	Available for an application-specific sensor
4	AN-4	Process temperature	Input	e.g. supply temperature from a thermal circuit
5	AN-3	Process temperature	Input	e.g. return temperature from a thermal circuit
6	AN-2	Free analog input	Input / output	Available for an application-specific sensor
7	AN-1	Free analog input	Input / output	Available for an application-specific sensor
8	AN-0	Free analog input	Input / output	Available for an application-specific sensor

4.2 Digital inputs and outputs

Digital sensors and simple on/off signals are connected to the board's I/O channels.

Signal	Channel	Generic function	Type	Typical use / notes
9–13	IO-7 ... IO-3	Free digital I/O	Input / output	Available for application-specific status signals or on/off outputs
14	IO-2	Alarm output	Output	General alarm signal to an external system
15	IO-1	Alarm / status input	Input	e.g. fault or ready signal from connected equipment (inverter, boiler, pump)
16	IO-0	Free digital I/O	Input / output	Available for an application-specific signal

4.3 Power / actuator channels

Higher-current and actuator-drive channels are available directly from the board and, where more channels are needed, from an extension board on the same bus.

Signal	Channel	Generic function	Type	Typical use / notes
17–18	JP18–JP19	Actuator control	Output	Position actuator or similar high-current output

Signal	Channel	Generic function	Type	Typical use / notes
19–22	JP10–JP7	Setpoint / analog output	Output	Analog setpoint to a connected device (e.g. inverter or valve actuator)
23–26	JP6–JP3	Switched output	Output	Pump, valve solenoid, contactor or similar switched load

Unused channels are available for application-specific extension without affecting the platform-level behaviour described in the rest of this manual.

5. Communication and Data Logging

The Melacs board logs process data to a removable microSD card and exposes the same data locally and over a network. This is independent of the connected application: whatever sensors and actuators are configured, their values are logged and made available through the same set of interfaces.



Figure 2 — Melacs microSD card and adapter

5.1 Local data logging

Process values are written to a dated log file on the memory card (for example, one file per day). Logged values typically include date, time, the active analog and digital channel values, board temperature and supply voltage, in addition to any application-specific derived values (such as calculated available energy).

The log data can be accessed in three ways:

- Connect a USB cable between the Melacs board and a PC — the board presents the memory card as a USB storage device (recommended).
- Remove the memory card and read it directly with a card reader on a PC.
- View current data and trends over the network via the board's built-in web server.

5.2 Network and fieldbus interfaces

- Ethernet — built-in web server for live values, trends and (where enabled) remote configuration.
- RS-485 / Modbus RTU — for integrating the Melacs board with other controllers, inverters or a site-level energy-management / SCADA system.
- USB — local access to the memory card and, where supported, firmware update / commissioning.
- Cellular (GSM), where fitted — remote connectivity for unmanned or distributed sites.

These interfaces allow a Melacs installation to operate either as a stand-alone controller or as one node in a larger, multi-site energy platform, with the same logged data set available locally and remotely.

6. Operating Modes

6.1 Normal operation

In normal operation the application software continuously reads the configured inputs, runs the application-specific control algorithm, and drives the configured outputs. The local display, where fitted, shows the primary process value(s) together with time and date, and provides a navigation menu to further pages of live data and settings.

Typical pages available during normal operation are:

- Primary process page — the main measured value(s) for the installation (for example, a controlled temperature).
- Secondary process pages — additional measured values such as flow, position or auxiliary sensors.
- System information page — internal Melacs diagnostic information (board temperature, supply voltage, firmware version).
- Settings menu — clock, A/D calibration, logging, network, and application-specific parameters.

6.2 Safe mode

If a monitored process value moves outside its configured safe range (for example, a temperature limit is reached or exceeded), the system enters a safe mode appropriate to the application: outputs are driven to a defined safe state (for example, an actuator is moved out of focus, a pump is stopped, or a valve is set to a fail-safe position) and an alarm condition is raised. The system returns to normal operation automatically once the process value has returned within the configured margin, unless the application requires manual acknowledgement.

6.3 Manual mode

Manual mode allows an operator to directly command outputs (for example, to run an actuator or a pump) for commissioning, maintenance or fault-finding, overriding the automatic control algorithm. It is recommended to isolate or de-energise the equipment being worked on before using manual mode for maintenance, and to return the system to normal operation afterwards.

6.4 Application-specific modes

Individual applications may define additional modes on top of the platform-level modes above — for example, a frost-protection mode, a forced-charging mode for a thermal or electrical store, or a grid-support mode for an inverter application. These are documented in the application supplement for the installation concerned.

7. Status and Alarm Codes

The Melacs platform reports a small set of generic status conditions that are common across applications; combinations are reported by adding the relevant codes, as shown below. Applications may extend this set with additional, application-specific codes documented in the relevant application supplement.

Code	Meaning
CODE 0	System OK — no active condition
CODE 1	A monitored value is above/below its configured limit
CODE 2	Warning reported by connected equipment (e.g. inverter, boiler, heat pump)
CODE 3	CODE 1 and CODE 2 active simultaneously
CODE 4	An actuator or mechanism has reached an end stop / limit switch
CODE 5	CODE 1 and CODE 4 active simultaneously

8. Typical Application Areas

Because the Melacs platform separates the generic board and firmware from the application-specific control program, the same hardware family has been applied across a range of energy plant types, including:

- Solar thermal systems — non-tracking and tracking collector fields, controlling collector pumps and valves from irradiance and temperature measurements.
- Heat pump installations — sequencing and monitoring of heat-pump operation alongside other heat sources.
- Thermal energy storage — charge/discharge control of accumulator tanks and stratified storage.
- Biofuel / biomass boilers — supervisory control and status monitoring of combustion-based heat sources within a wider system.
- Power electronics and grid interfaces — logging and supervisory control around grid inverters and Energy-as-a-Service (EaaS) functions.
- Power-to-X and e-fuel systems — process monitoring and control around electrolysis and downstream fuel-synthesis equipment.
- Geothermal and district heating/cooling interfaces — monitoring and control of ground-source and district network connections.

This list illustrates the breadth of application rather than limiting it: any plant that can be described in terms of analog/digital sensing, actuator control, logging and communication can, in principle, be implemented on the Melacs platform through its application layer.

9. Commissioning and Safety Notes

- Verify sensor and actuator wiring against the application supplement's channel list before applying power.
- Set the correct date and time before relying on logged data; the real-time clock retains time for a limited period without power.
- Confirm that safe-mode limits (e.g. maximum temperature) are set correctly for the connected equipment before leaving the system unattended.
- Use manual mode only for commissioning and maintenance, with the relevant equipment isolated where appropriate, and return the system to normal (automatic) operation afterwards.

- Keep the memory card log for the commissioning period to assist with any later performance review or fault diagnosis.
- Refer to the application supplement for any electrical safety, pressure or process-specific precautions relevant to the connected plant.

10. Document Revision History

Revision	Date	Description
Rev. A	June 2011	Original application-specific example manual (solar-tracking heating system), Finsun Inresol AB.
1.0	2026	General edition: application-specific content replaced with a generic description of the Melacs control platform (architecture, I/O, communication, operating modes, status codes) applicable to any connected energy plant.

Melacs™ is a trademark used in connection with the Melacs control platform. Product and platform details are subject to change as development continues; consult the current application supplement for installation-specific information.